Perceptron geometry

Take input $x \neq 0$ with magnitude r and direction ϕ, weight w with magnitude ρ and direction θ, and bias b.

$$\mathbf{x} = r\mathbf{v}, \qquad \mathbf{w} = \rho\mathbf{u}, \qquad \text{where} \quad r = \|\mathbf{x}\|, \qquad \rho = \|\mathbf{w}\|, \qquad \|\mathbf{v}\| = \|\mathbf{u}\| = 1.$$

Now we have, $\mathbf{u} = (\cos\theta, \sin\theta), \mathbf{v} = (\cos\phi, \sin\phi)$ with score

$$f_{\theta,\rho,b}(\mathbf{x}) = \rho r \cos(\phi - \theta) + b$$

by dot product. Perceptron classification gives

$$h(\mathbf{x}) = 1 \iff r\cos\alpha \geq -\frac{b}{\rho}$$

with decision boundary $\in \mathbb{R}$

$$\mathcal{H} = \{\mathbf{x} : \mathbf{w}^\mathsf{T}\mathbf{x} + b = 0\}.$$

Its normal direction is $\mathbf{u}$, and its signed distance from an input is

$$d(\mathbf{x}) = \frac{f(\mathbf{x})}{\rho} = r\cos\alpha + \frac{b}{\rho}.$$

Weight normalization preserves the classifier when the bias is rescaled too. Divide the entire score by $\rho > 0$:

$$\widetilde{f}(\mathbf{x}) = \mathbf{u}^\mathsf{T}\mathbf{x} + \widetilde{b}, \qquad \widetilde{b} = \frac{b}{\rho}.$$

The score's sign is unchanged. We can therefore work with unit weights and write, using b for the rescaled bias,

$$f(\mathbf{x}) = r\cos\alpha + b.$$

Input normalization is a separate modeling choice. Replacing $\mathbf{x}$ by $\mathbf{v} = \mathbf{x}/r$ gives

$$f(\mathbf{v}) = \cos\alpha + b.$$

This removes magnitude information. With zero bias, classification depends only on angle:

$$h(\mathbf{x}) = 1 \iff \alpha \leq \frac{\pi}{2}.$$

With nonzero bias, magnitude can matter. For example, an input aligned with the weights has $\alpha = 0$, so with unit weights and $b = -2$,

$$f(\mathbf{x}) = r - 2.$$

An input of length 3 receives class 1; normalizing it to length 1 changes its class to 0. If both weights and inputs are unit vectors, then for $-1 < b < 1$,

$$h(\mathbf{x}) = 1 \iff \alpha \leq \arccos(-b).$$

Thus the bias sets an angular cutoff on the unit sphere. Without input normalization, that cutoff generally depends on r. The calculus is ordinary real multivariable calculus. In Cartesian coordinates,

$$\nabla_{\mathbf{x}} f = \mathbf{w}.$$

Assuming $\mathbf{w} \neq \mathbf{0}$ gives the boundary regularity condition

$$f(\mathbf{x}) = 0 \quad \Longrightarrow \quad \nabla_{\mathbf{x}} f(\mathbf{x}) \neq \mathbf{0}.$$

The score is smooth; thresholding it produces a classifier discontinuous at the boundary. In the two-dimensional angular parametrization,

$$f = \rho r \cos(\phi - \theta) + b,$$

and, holding the input fixed,

$$\frac{\partial f}{\partial \theta} = \rho r \sin(\phi - \theta), \qquad \frac{\partial f}{\partial \rho} = r \cos(\phi - \theta), \qquad \frac{\partial f}{\partial b} = 1.$$

For a differentiable training loss $\ell(f, y)$, the chain rule gives

$$\frac{\partial \ell}{\partial \theta} = \frac{\partial \ell}{\partial f} \rho r \sin(\phi - \theta).$$

The perceptron predicts by thresholding:

$$h(\mathbf{x}) = \mathbf{1}_{\{f(\mathbf{x}) \geq 0\}},$$

and is not everywhere continuous. Apply a loss directly to the smooth score $f(\mathbf{x})$. For convenience, encode labels as $y \in -1, +1$. Define the signed margin

$$m = y f(\mathbf{x}).$$

A positive margin means the score has the correct sign; a negative margin means the wrong sign. At $m = 0$, the prediction depends on our boundary convention. A common differentiable choice is the logistic loss:

$$\ell(f, y) = \log\left(1 + e^{-yf}\right).$$

It decreases as the signed margin increases. Its derivative with respect to the score is

$$\frac{\partial \ell}{\partial f} = -\frac{y}{1 + e^{yf}}.$$

For $y = +1$, this derivative is negative, so reducing the loss encourages a larger score. For $y = -1$, it is positive, so reducing the loss encourages a smaller score. Using our angular parametrization,

$$f = \rho r \cos(\phi - \theta) + b,$$

the chain rule gives

$$\boxed{\frac{\partial \ell}{\partial \theta} = -\frac{y\,\rho r \sin(\phi - \theta)}{1 + e^{yf}}}.$$

Gradient descent updates the direction by

$$\theta_{\text{new}} = \theta - \eta \frac{\partial \ell}{\partial \theta}, \qquad \eta > 0.$$

For example, suppose $y = +1$ and the input direction is slightly counterclockwise from the weight direction:

$$0 < \phi - \theta < \frac{\pi}{2}.$$

Then $\sin(\phi - \theta) > 0$, so

$$\frac{\partial \ell}{\partial \theta} < 0.$$

The update increases θ, rotating the weights toward that input and increasing its score. The other parameter derivatives are

$$\frac{\partial \ell}{\partial \rho} = -\frac{y\, r \cos(\phi - \theta)}{1 + e^{yf}}, \qquad \frac{\partial \ell}{\partial b} = -\frac{y}{1 + e^{yf}}.$$

These describe changes to weight magnitude and boundary position. For a dataset $(\mathbf{x}_i, y_i)_{i=1}^N$, we minimise the average loss:

$$L(\theta, \rho, b) = \frac{1}{N} \sum_{i=1}^{N} \log\left(1 + e^{-y_i f(\mathbf{x}_i)}\right).$$

Its gradient averages the contributions from all examples, so training balances their competing effects. One useful distinction from our earlier discussion: with zero bias, magnitude does not affect the predicted class, but it still affects the loss and training gradients. Two correctly classified inputs at the same angle can have different margins and therefore different losses. In Cartesian weights, logistic loss is convex in $(\mathbf{w}, b)$. The angular parametrization makes rotations easier to interpret, but the resulting objective need not be convex in θ. This is a reason to use angles for understanding the geometry and Cartesian weights for analyzing optimization.